Quiz 8 —

(1) $y^2 = 1-x$ and $2y = x+2$

$1-x = y^2 \Rightarrow \boxed{x = 1-y^2}$

$2y = x+2 \Rightarrow \boxed{x = 2y-2}$

$1-y^2 = 2y-2 \Rightarrow y^2+2y-2-1 \Rightarrow y^2+2y-3=0$

$(y+3)(y-1) = 0 \Rightarrow \boxed{y=-3} \quad \boxed{y=1}$

$$A = \int_{-3}^{1} (\text{Top} - \text{bottom})\, dy$$

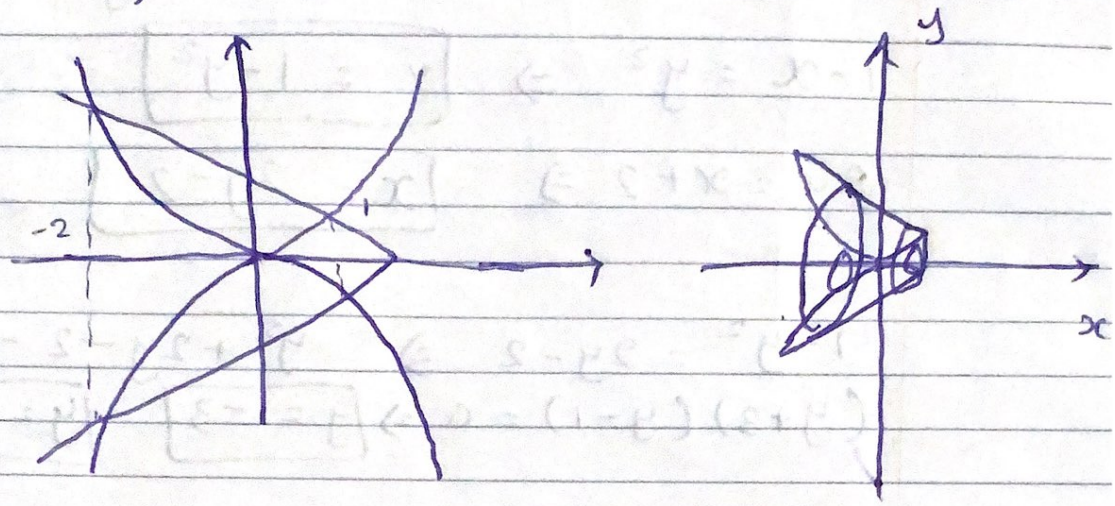
$y=0 \Rightarrow 1-0=1$, $y=0 \Rightarrow -2$

$$A = \int_{-3}^{1} [(1-y^2) - (2y-2)]\, dy$$

$$A = \int_{-3}^{1} (1-y^2-2y+2)\, dy$$

$$\boxed{A = \int_{-3}^{1} -(y^2+2y-3)\, dy}$$

② $y = x^2$, $x + y = 2$ and $x = 0$



$$V = \pi \int_a^b \left[(\text{outer})^2 - (\text{inner})^2 \right] dx$$

$$V = \pi \int_{-2}^{1} \left[(2-x)^2 - (x^2)^2 \right] dx$$

Check :-

$y = 2 - x \Rightarrow x^2 = 2 - x \Rightarrow x^2 + x - 2 = 0$

$(x-1)(x+2) = 0 \Rightarrow \boxed{x = 1} \quad \boxed{x = -2}$

$x = 0 \Rightarrow$

$y = x^2 \Rightarrow y = 0$, $y = 2$ on $x + y = 2$

$$V = \pi \int_{-2}^{1} \left[(2 - 4x + x^2) - x^4 \right] dx$$

$$V = \pi \int_{-2}^{1} (x^2 - x^4 - 4x + 2)\, dx$$